Bitrate Adaptation

Samplerate –

1. A bitrate selection algo can't conclude that higher bitrates will perform poorly just because lower ones perform poorly.

Algo: –

1. Samplerate sends data at the highest possible bitrate
2. It stops a bitrate if it experiences 4 succ. failures
3. Samplerate when it experiences failures it will keep lowering bitrate until it finds the appropriate
4. It samples a new bitrate at every 10th packet

Sample [: Two succ. failure, a bitrate that provides average lower lossless time than the current avg time]

$$\text{Transmission time} = \frac{\text{Packet size}}{\text{Bitrate}} \quad 11$$

5.5

if I am making n attempts before success

$$\text{avg. tx time} = (n+1) \times \frac{\text{Packet size}}{\text{bitrate}}$$

$$\text{Lossless Tx time} = \frac{\text{Packet size}}{\text{Bitrate}}$$

Dest	Bitrate	Succ. failure	Avg Tx	Lossless Tx
~~A~~	~~11~~	~~4~~	~~∞~~	~~1873~~
→ A	5.5	0	2976	2976
A	2	0	–	6834
A	1	0	–	12095

Samplerate will sample the 10th packet at a bitrate which will give lower lossless tx time < current avg tx time.

De

Dest	Bitrate	Succ. fail	Avg Tx	Lossless Tx
B	11	0	33761	1873
B	5.5	0	12000	2976
B	2	0	10990	6834
B	1	0	—	12555